

Local AI on Jetson Nano

Jetson Nano 4GB
Ubuntu 18.04.6 / JetPack 4
Python 3.11



Setup

Run in a terminal on the Nano.

```
git clone https://github.com/nileshsarkar-ai/JetsonNano-Demo.git
```

```
cd JetsonNano-Demo
```

```
bash scripts/run_demo.sh --storage
```

```
bash scripts/run_demo.sh --check
```

```
bash scripts/run_demo.sh --setup-only
```

```
bash scripts/run_demo.sh
```

Text Conversation

Menu 3

“Why does a steel ship float?”

Follow-up prompts

“Explain it using an everyday analogy.”

“Now explain it in two sentences.”

SmolLM2-360M Instruct uses the preceding conversation.

/reset clears history. /quit returns to the menu.

Camera Object Detection

Menu 4

SSD-Mobilenet-v2

Place everyday objects
in the camera's view.

Move one object.
Turn the book sideways.
Partly hide the cup.



Illustrative scene

Camera-Guided Object Hunt

Menu 5

- 1 The language model proposes objects to find.
- 2 Arrange the objects in view and press Enter to scan.
- 3 Camera detections update the found and remaining lists.

The hunt shares the conversation model and object detector.

Models and Execution

Text

SmolLM2-360M Q4
258 MiB model file
CPU inference

Camera

SSD-Mobilenet-v2
TensorRT on the Nano GPU
Camera and text stages run sequentially

Menu Controls

- 1** Prepare models and dependencies
- 2** Check board status
- 3 / 4 / 5** Run a demonstration
- 6** Inspect storage usage
- 7** Open this presentation
- 0** Exit

Ctrl+C cancels a demo and returns to the menu.